

Partha Chakraborty

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SUMMARY

Full-stack engineer with 2+ years at startups, building web and mobile products in React, Next.js, and TypeScript on Python, FastAPI, and Node.js backends. AI work includes LoRA fine-tuning of Qwen2.5-7B and code retrieval with hybrid semantic and BM25 search. Completing an M.S. in AI and Machine Learning at the University at Buffalo, expected January 2027.

EXPERIENCE

TurboMail

AI Engineer Intern

Remote

May 2026 – Aug 2026

- Owned production features across the stack in React, TypeScript, Python, FastAPI, SQLAlchemy, and SQL.
- Designed the database schema and API endpoints for subscription and payment tracking, then built the customer-facing React interface on top of them.
- Wrote bulk email deletion and folder-move APIs, plus email-body sanitization and thread scoping across multiple accounts.
- Cut down manual unsubscribe work with FastAPI background tasks that parse RFC 8058 headers.
- Fine-tuned Qwen2.5-7B with LoRA and PEFT in PyTorch on a custom email dataset to support product functionality.

Bird Vision AB

Software Developer

Remote

Jun 2023 – Jul 2025

- Ranked second in team code contributions within a year while shipping frontend and backend features across three startup products.
- Improved the Eboka customer interface with a team using React, Next.js, Redux, and Tailwind CSS, including responsive layouts for phone and desktop screens.
- Built and deployed a cross-platform employee app with React Native and Expo.
- Wrote Cypress tests covering key user flows in the web app.
- Integrated third-party services into backend workflows with Node.js, Express.js, and REST APIs, and handled production deployment for my features.

EDUCATION

University at Buffalo

M.S. in Artificial Intelligence and Machine Learning

New York, USA

GPA 3.9/4.0 | Expected Jan 2027

Coursework: Algorithms, Database Systems, Data-Intensive Computing, Machine Learning

Presidency University

B.Tech. in Computer Science and Engineering

Bengaluru, India

GPA 8.14/10 | 2018 – 2022

Coursework: Computer Networks

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, HTML, CSS

Frontend: React, Next.js, Redux, Tailwind CSS, responsive design, React Native, Expo, Cypress

Backend: FastAPI, Node.js, Express.js, REST APIs, HTTP, SQLAlchemy, background tasks, third-party API integrations

AI and ML: PyTorch, LoRA, PEFT, RAG, embeddings, vector search, BM25, NumPy, NetworkX

Data and tools: PostgreSQL, MySQL, MongoDB, Firebase, ChromaDB, Qdrant, Docker, Linux, Git, Streamlit, Altair

AI coding tools: Cursor, Claude Code, GitHub Copilot

PROJECTS

AI Codebase Intelligence Platform (in progress) [GitHub]

Python, ChromaDB, Qdrant

- Built repository ingestion for six languages (Python, JavaScript, TypeScript, Java, C++, Go), with language-aware chunking and metadata extraction for vector indexing.
- Combines semantic and BM25 retrieval to pick the relevant code chunks for a natural-language question, so the full repository never goes to the model.

GraphBench [Live Demo]

Python, NetworkX, LadybugDB, Streamlit, Altair

- Built a dependency analysis tool using PageRank, betweenness centrality, cycle detection, and K-core analysis to find critical dependencies, bottlenecks, and cycles.
- Guided CSV ingestion validates data before loading; an interactive explorer and JSON/CSV reports explain findings in plain language.
- A failure-impact simulation traces direct and indirect dependents, calculates blast radius and impact depth, and shows the shortest dependency paths.

Isolation Forest [Live Demo]

Python, FastAPI, Next.js, NumPy

- Implemented the algorithm from its original paper: recursive tree construction, training, and anomaly scoring.
- Wrapped it in a FastAPI backend and Next.js dashboard for uploading data, running the model, and exploring anomalies.